

TED (15)-3253

Reg.No.....

(REVISION 2015)

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY (CABM)

(Model question paper)

QUANTITATIVE TECHNIQUES

[Time: 3 hours]

(Maximum marks: 100)

PART A

I (Answer the following questions in one or two sentences. Each question carries 2 marks)

1. Outline the primary data.
2. State the term median.
3. Quote the co-efficient of correlation.
4. Recite the meaning of consumer price index.
5. List out any two limitations of statistics.

(5*2=10)

PART B

II (Answer any five questions, each question carries 6 mark)

1. Recognise the importance of statistics.
2. List out the disadvantages of mode.
3. Calculate Pearson's co-efficient of correlation of the given values.

X: 2 3 4 5 6 7 8

Y: 4 5 6 12 9 5 4

4. Production of Malabar foods in India is as following for two years. Calculate index number for 1991 and 1992 taking 1991 as base year.

Grains	Quantity	
	1991	1992
Wheat	4950	5040
Maize	3060	4080
Barley	1024	1526
Paddy	2460	3040

5. State the objectives of classification.
6. List out the disadvantages of median.
7. Illustrate the problems in construction of index numbers.

(5*6=30)

PART C

(Answer one full question from each module, each full question carries 15 marks)

(4*15=60)

MODULE - I

- III. (a) State the different types of data. (8)
- (b) Distinguish between primary data and secondary data. (7)

Or

- IV. Memorise classification and its types. (15)

MODULE - II

- V. Find the quartile deviation and its co-efficient from the following data (15)

Age(in years)	15	16	17	18	19	20	21
No. of students	4	6	10	15	12	9	4

Or

VI. Calculate weighted arithmetic average from the following data. (15)

Value	60	48	55	72	45
Weight	2	1	3	4	2

MODULE - III

VII. Calculate spearman's correlation co-efficient. (15)

X	40	45	50	43	40	75	55	72	65	42	70
Y	40	46	54	60	70	80	82	85	85	90	90

Or

VIII. Calculate correlation co-efficient between X and Y. (15)

X	10	20	30	40	50	60	70	80	90	100
Y	2	4	8	5	10	15	14	20	22	30

MODULE - IV

IX. Explain in detail the problem in constructing index numbers. (15)

Or

X. From the following data construct index numbers. (15)

- (a) Laspeyre's formulae
- (b) Paasche's formulae
- (c) Fisher's formulae
- (d) Marshall-Edgeworth formulae

Commodity	Prices in (Rs.)		Quantity	
	2015	2016	2015	2016
A	10	15	10	8
B	50	60	12	10
C	80	100	50	20
